

SkyBook

An airline reservations platform, built end to end

Recruiter edition — the 12-page summary of a 68-page engineering case study

At a glance

Item	Detail
What	A working airline: anonymous search, real seat maps, payment, confirmation email, check-in, scannable boarding pass, cancellations with fare-rule refunds, an admin console
Shape	Eight Spring Boot applications (one API gateway, seven domain services), React 19 web client, PostgreSQL, Kafka, full observability
Built	35 calendar days of designed increments, then a measured quality and release-engineering revision
Quality	SonarCloud gate PASSING: 91.2% coverage, 0 bugs, 0 vulnerabilities, security and reliability rating A
Tests	1,722 backend executions + 55 frontend + a 24-test end-to-end certification that races real buyers for one seat - zero failures at the evidence commit
Environments	A real promotion ladder: LOCAL > DEV > SIT > QA > PERF > UAT > STAGING > PROD, plus a weekly disaster-recovery drill that restores and verifies row counts
Deployment	One cloud VM, 18 containers, automatic TLS - promoted by image digest, never rebuilt on the way to production
Role	Solo: requirements, architecture, backend, frontend, testing, CI/CD, cloud, documentation. Built using AI-assisted development tooling, with every decision and merge owned by the author

[Repository: github.com/praveencloudlab/skybook](https://github.com/praveencloudlab/skybook)

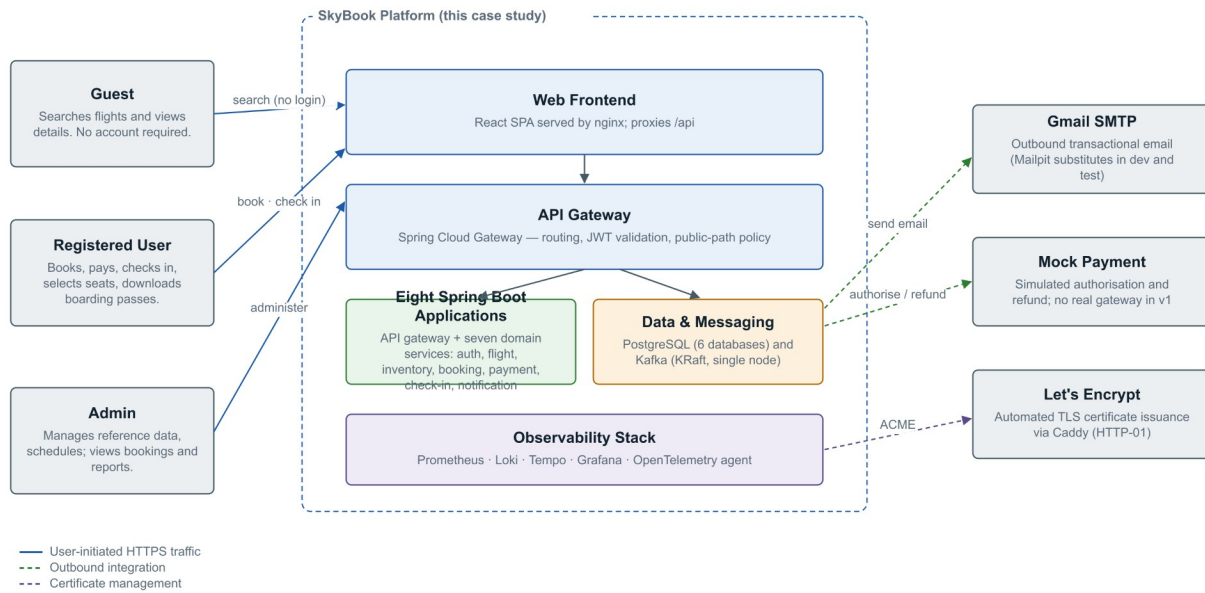
[Quality dashboard: sonarcloud.io/project/overview?id=praveencloudlab_skybook](https://sonarcloud.io/project/overview?id=praveencloudlab_skybook)

Disclaimer: SkyBook is a personal portfolio project - not a real airline, affiliated with no carrier. Airline-style codes in seed data are illustration only; payments are simulated server-side and no cardholder data exists; all passenger records are synthetic.

The system

Figure 1 — System Context

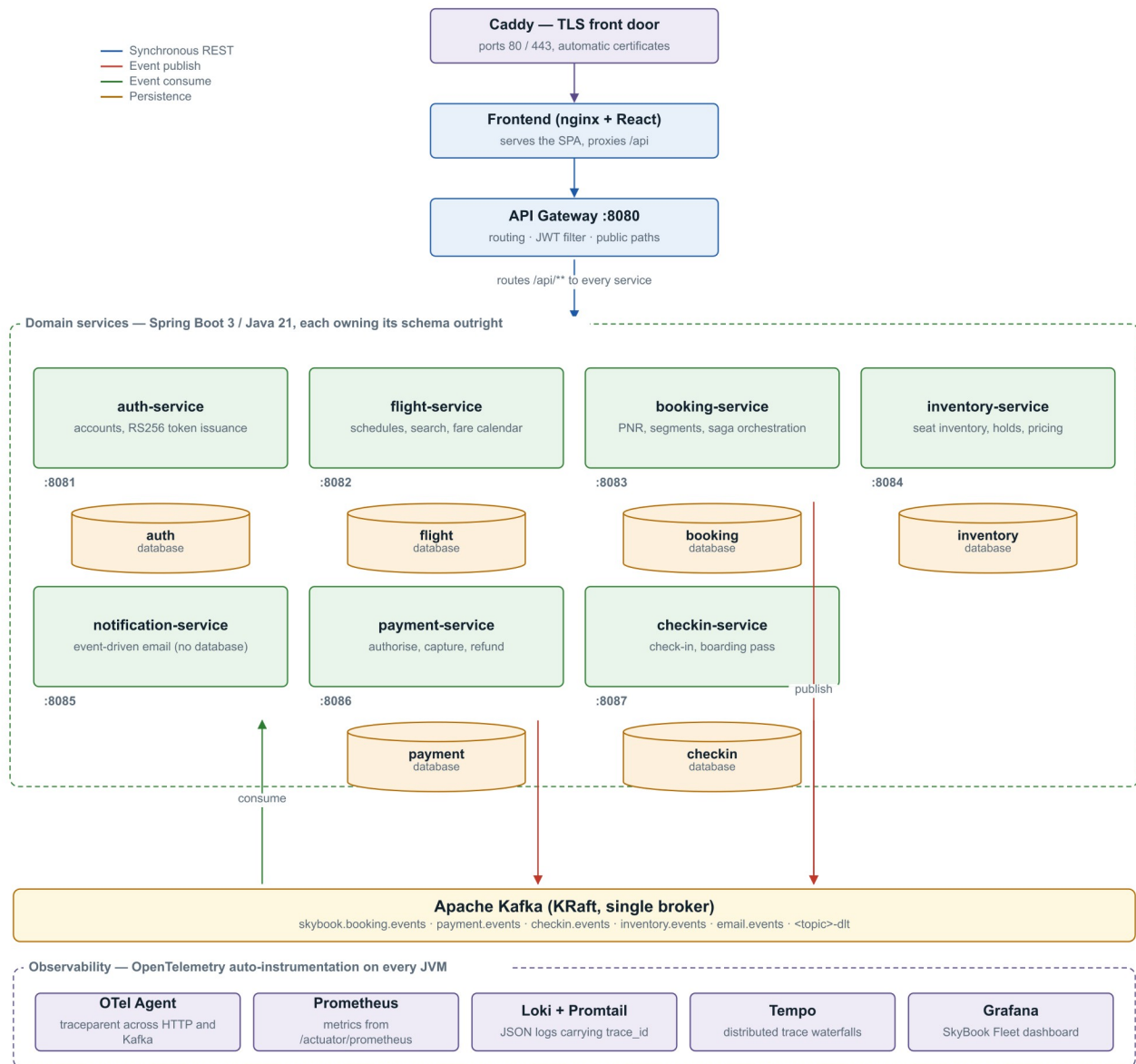
SkyBook passenger reservation platform and the parties around it



The context: a public shopping surface, an authenticated booking core, and operations behind an admin role.

Figure 2 — Microservice Architecture

Runtime components, synchronous routes and asynchronous event flow

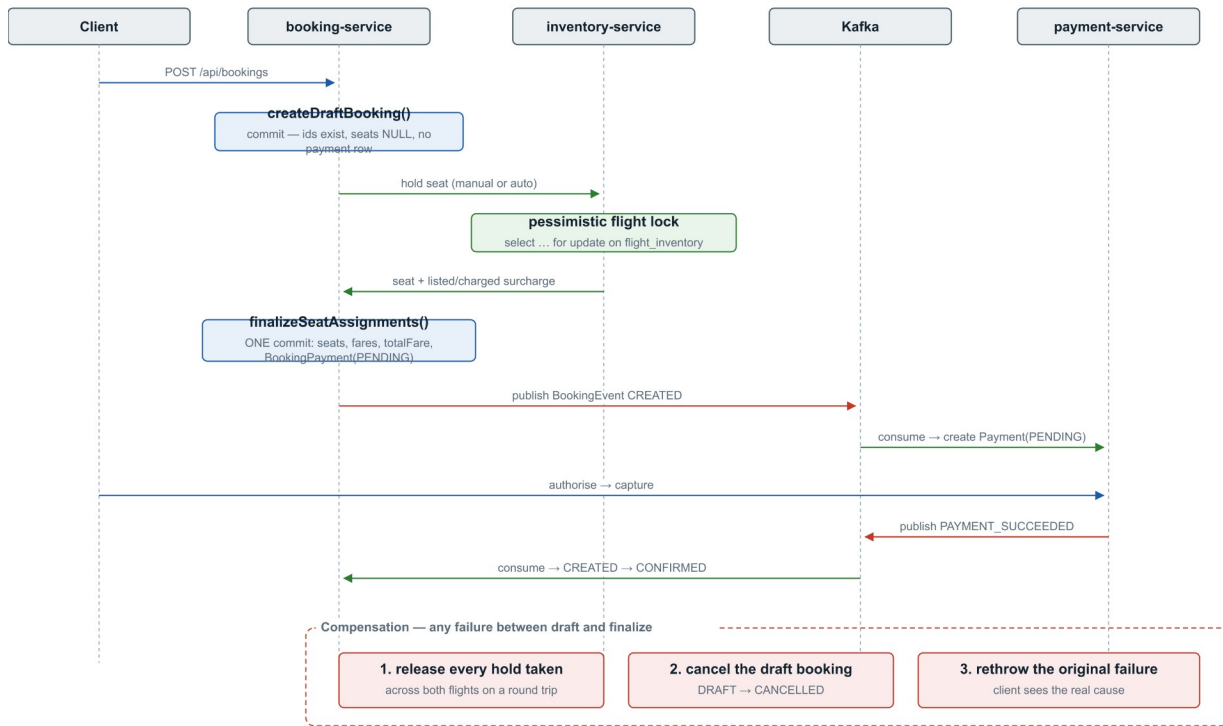


Eight applications over one database instance (six schemas) and one event bus. Every service owns its data; nothing reaches another service's tables.

The hard part: money moves asynchronously

A booking is not confirmed when the payment call returns - it is confirmed when payment-service says so over Kafka and booking-service reacts. The saga below is the platform's spine, and the double-sell race (two real buyers, one seat, concurrent checkout) is certified end to end every night and on every promotion.

Figure 5 — Booking Saga
draft → hold → finalize, with compensation on any failure

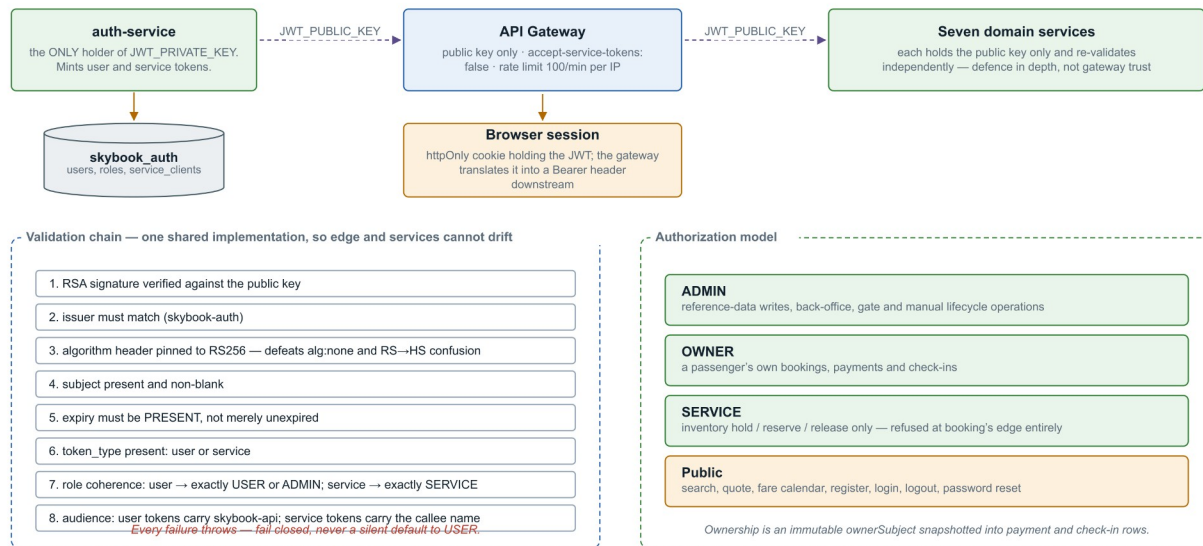


The facade is deliberately NOT @Transactional: every external call happens outside a database transaction, so no connection is held across network I/O.

The booking saga: draft, hold, pay, confirm - with compensations, not distributed locks.

Security, stated honestly

Figure 10 — Authentication and Authorization
RS256 asymmetric signing: one service mints, everyone verifies



RS256 sessions, per-service credentials, an httpOnly cookie the gateway exchanges for a Bearer header - and the gaps stated next to the controls.

The eight CSRF findings SonarCloud raised were closed with written justifications in code - the cookie is consumed at the gateway, SameSite=Lax covers the SPA, and all 48 GET endpoints were audited as read-only. A leaked mail credential early in the project was revoked, purged from history, and is recorded in the study as an incident, not hidden.

Evidence, not adjectives

Claim	Proof
1,722 tests, zero failures	mvn clean verify at commit 96e24b2, 4 Aug 2026 - per-module table in the full study sums to exactly this
91.2% coverage, gate passing	Public SonarCloud dashboard, measured on every push
The journey certifies	24 e2e tests through the gateway only: booking, payment, check-in, boarding pass, double-sell race, email into a real sink, one trace across the Kafka hop
Fast enough under load	k6: 3,405/3,405 checks, p95 search 60 ms, quote 26 ms, 0.00% errors - thresholds gate every promotion
Backups actually restore	Weekly drill: back up, destroy the database volume, restore, verify 1,336,748 rows against the manifest - drilled live on 4 Aug
Same artifact everywhere	Images built once per commit, scanned, published as multi-arch digests; every environment from DEV to PROD runs those exact bytes

The product

The screenshot displays the SkyBook flight search interface. At the top, the SkyBook logo is on the left, and language (English), currency (£ GBP), and search/sign-in buttons are on the right. The main heading is "Search flights" with a subtext: "A year of real schedules across 30 routes. Compare fares, pick your seat from the actual cabin — no account needed to look." Below this is a search bar with tabs for "Round trip", "One way" (selected), and "Multi-city". The search criteria are: From London Heathrow (LHR), To Dubai (DXB), 1 Guest, Economy, Travelling when? Tue 4 Aug. A "Search" button is to the right.

Below the search bar, a seven-day price strip shows prices for August 1st to 7th. The prices are: 1 Aug —, 2 Aug —, 3 Aug £136, 4 Aug £136 (highlighted), 5 Aug £136, 6 Aug £136, 7 Aug £130.90.

The main results area shows "20 trips" for the route "LHR — DXB · Tue 4 Aug". On the left, there are filters for "STOPS" (Direct: 4, 1 stop: 15, 2 stops: 1) and "DEPARTURE TIME" (Morning: 05:00 – 11:59, Afternoon: 12:00 – 16:59, Evening: 17:00 – 20:59). The "SORT BY" options are Earliest, Latest, and Shortest.

The flight results are listed as follows:

Flight Type	Departure (LHR)	Arrival (DXB)	Duration	Fare (per person)
Direct	08:25	15:20	6h 55m	£136.00
Direct	06:53	13:53	7h	£136.00
Direct	13:29	20:29	7h	£136.00

Search: a seven-day price strip and one card per itinerary - priced by the same endpoint checkout will use.

English
GBP
Search flights
My trips
Profile
Sign out

Booking in progress: LHR → DXB 2026-08-05
Start a new search

LHR → DXB
Wed 05 Aug 08:25
Back

Flights
Guests
Seats
Bags
Payment

Seat selection

Economy - Saver — secure your seat now, or let us assign one free at check-in.

Case Study

☐ Free
☐ Extra charge
☒ Selected
☐ Unavailable
☐ Exit row

FRONT OF AIRCRAFT

BEAR OF AIRCRAFT

No seat chosen Skip — assign me a seat (free)

Back Add seats later Continue

Summary

[Flight details](#)

Wed 05 Aug

08:25 → **18:20**

LHR → DXB

6h 55m Direct

Economy Saver

London Heathrow to Dubai

Guests

Mr Case Study

Fare summary

Total: **£136.00**

Including taxes:

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A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

BOOK

Search flights

My trips

ACCOUNT

Signed in as [passenger.de...](#)

Sign out

© 2026 SkyBook. A portfolio project — not a real airline.

30 routes · 16 airports · a year of departures

Seat selection from the real cabin: paid rows priced per seat, held seats disabled, free auto-assign.

 SkyBook

English ▼ £ GBP ▼ Search flights My trips Profile Sign out



Payment received

Your booking reference

SB4EFA

Paid£136.00

Payment referencePAY-2026-T887GQ

Seat19B

StatusCONFIRMED

E-ticket

Case Study125-0000023201

Confirmed. A confirmation email is on its way.

Done



A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

BOOK

Search flights
My trips

ACCOUNT

Signed in as **passenger.de...**
Sign out

The moment three services agree: record locator, paid amount, trip saved.

All trips

BOOKING REFERENCE
SB4EFA
Booked Tue 4 Aug

Confirmed

£136.00
Download e-ticket

EK EK0079 Checked in

05:45

6h 50m
Direct

09:35
MAN

Passenger (1)

Seats 19B

Case Study
Economy - Saver - seat 19B

£136.00

Total paid

£136.00

CONTACT

NAME
Case Study

EMAIL
passenger.demo@skybook.test

PHONE
Not provided at booking

SkyBook

English

£ GBP

Search flights

My trips

Profile

Sign out

Case Study

125-0000023201 CT CHECKED_IN

Manage booking

Cancellation & refund rules

- More than 72h before departure — full refund per your fare rules (Saver keeps its 30% fee; Flexi and Premium refund in full).
- 72h – 24h before departure — 50% of the fare-rule refund.
- Under 24h (same day) — cancellation still frees your seat, but the fare is not refunded.
- Last 2 hours before departure — online cancellation is closed; the airport desk can still help.
- Already checked in? — you can still cancel the whole booking; issued boarding passes are voided and the seats released.
- A refund due is returned automatically to the original payment method; check-in closes for every passenger.

Cancel booking...

Check-in & boarding passes

Case Study

EK0079 DXB → MAN Wed 5 Aug, 05:45 Seat 19B

Change seat checked in

Download boarding pass

SkyBook

BOARDING PASS

ECONOMY

PASSENGER
CASE STUDY

BOOKING REF (PNR)
SB4EFA

DXB
Dubai - Terminal 3

MAN
Manchester - Terminal 2

FLIGHT
EK0079

DATE
5 Aug 2026

DEPARTS
05:45

BOARDING
05:00

TERMINAL
3

GATE
TBA

SEAT
19B

CABIN
Economy

BOARDING GROUP
4

NOTICE: Please arrive at the boarding gate by 04:30, 30 minutes before boarding begins at 05:00. The gate closes before departure and late passengers may be offloaded.

Issued 4 Aug 2026 11:04

BOARDING PASS

PASSENGER
CASE STUDY

FLIGHT
EK0079

DXB → MAN

SEAT
19B

GSP
4

BOARDS
05:00

QR CODE

BP-2026-KPTGBR
PNR SB4EFA

SkyBook

A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

BOOK

Search flights

My trips

ACCOUNT

Signed in as passenger.de...

Sign out

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30 routes · 16 airports · a year of departures

The boarding pass - the QR is real and the gate-operations console verifies it.

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Five things that went wrong, on purpose in print

Incident	Lesson
A transactional method silently ran with no transaction (Spring self-invocation)	Proxies do not apply to internal calls; the nightly schedule generator ran unprotected until static analysis flagged it
Every cross-timezone arrival displayed hours wrong - durations looked right by accident	Two wall clocks in different zones subtract to nonsense; fixing it touched generation, email, search, 414,679 rows and the screenshots
A round-trip cancellation could refund partially and leave the booking looking live	The expansion set and the remaining set were computed from different keys; pinned by tests that fail on the old code
The environment ladder's first ever run refused a bad promotion, then caught its own seeding bug	Gates that only pass are decoration; these blocked, and the fixes are commits
A test demanded zero collisions from a generator whose own name promised "rare"	The birthday paradox made CI red one run in twenty-two; assert what the code promises, not more

Current limits, stated: no transactional outbox (a post-commit publish failure loses the event - top of the roadmap), no sustained capacity test (the k6 gate catches regressions, not limits), single-VM deployment with RPO 24h / RTO ~30min, and a browser UI suite not yet wired into CI. The full study gives every gap its reasons.

The full 68-page engineering edition covers the architecture, the saga, pricing and refunds, check-in, the delivery history commit by commit, testing, defects and lessons, the data model, and six appendices - traceability, API contracts, security operations, screens, roadmap, glossary.